

Infrastructure Design Document

Servers, networks, storage and platform services

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1. Introduction

This document describes the technology infrastructure that underpins the ArchiSurance application portfolio. It covers compute nodes, system software, network topology, storage architecture, and deployment artifacts. For the application-level view, refer to the Application Portfolio and Inventory (EA-APP-002). The Disaster Recovery and Business Continuity Plan (EA-DR-010) details failover procedures for the infrastructure described herein.

Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Integration testing spans three environments: development, staging, and pre-production. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Lessons learned from the pilot phase have been incorporated into the current design iteration. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern.

A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The incident management process defines four severity levels, with Severity

1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes.

The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys.

Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Change management activities include stakeholder communication, training sessions, and documentation updates. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner.

Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard

shortcuts for power-user workflows. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager.

2. Compute Infrastructure

ArchiSurance operates ten compute nodes across two data centers and one cloud region:

App Server Cluster

Hosts: K8s, the container runtime, Red Hat Linux 9.

The App Nodes is located in the primary Amsterdam data center (AMS-DC1). It consists of 16 vCPU cores and 64 GB RAM. Hardware refresh is scheduled for Q2 2026 as part of the infrastructure modernization program.

From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Security review was completed in accordance with the CISO's security-by-design framework. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates.

The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Integration testing spans three environments: development, staging, and pre-production. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations.

A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. Integration testing spans three environments: development, staging, and pre-production. The vendor selection process followed the procurement guidelines established by the CFO's office. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures.

Database Server Primary

Hosts: the primary relational database, Oracle DB.

The Database Server Primary is located in the primary Amsterdam data center (AMS-DC1). It consists of 8 vCPU cores and 32 GB RAM. Hardware refresh is scheduled for Q2 2026 as part of the infrastructure modernization program.

The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Lessons learned from the pilot phase have been incorporated into the current design iteration. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Change management activities include stakeholder communication, training sessions, and documentation updates. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer.

The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints.

Integration testing spans three environments: development, staging, and pre-production. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The vendor selection process followed the procurement guidelines established by the CFO's office. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer.

Database Server Replica

Hosts: PostgreSQL DB.

The Database Server Replica is located in the primary Amsterdam data center (AMS-DC1). It consists of 8 vCPU cores and 32 GB RAM. Hardware refresh is scheduled for Q2 2026 as part of the infrastructure modernization program.

An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Lessons learned from the pilot phase have been incorporated into the current design iteration. Change management activities include stakeholder communication, training sessions, and documentation

updates. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline.

The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025.

From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer.

Batch Processing Node

Hosts: Kafka.

The Batch Processing Node is located in the primary Amsterdam data center (AMS-DC1). It consists of 8 vCPU cores and 32 GB RAM. Hardware refresh is scheduled for Q2 2026 as part of the infrastructure modernization program.

Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Integration testing spans three environments: development, staging, and pre-production. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request.

Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The project governance structure

includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request.

The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days.

CI/CD Build Server

Hosts: Docker Engine.

The CI/CD Build Server is located in the primary Amsterdam data center (AMS-DC1). It consists of 8 vCPU cores and 32 GB RAM. Hardware refresh is scheduled for Q2 2026 as part of the infrastructure modernization program.

The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days.

The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations.

The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall

system availability or data integrity. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format.

Monitoring Server

Hosts: the search engine cluster.

The Monitoring Server is located in the primary Amsterdam data center (AMS-DC1). It consists of 8 vCPU cores and 32 GB RAM. Hardware refresh is scheduled for Q2 2026 as part of the infrastructure modernization program.

The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards.

The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity.

The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025.

Log Aggregation Node

Hosts: the search engine cluster.

The Log Aggregation Node is located in the primary Amsterdam data center (AMS-DC1). It consists of 8 vCPU cores and 32 GB RAM. Hardware refresh is scheduled for Q2 2026 as part of the infrastructure modernization program.

In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog.

Integration testing spans three environments: development, staging, and pre-production. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period.

Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices.

DR Recovery Node

Hosts: Postgres.

The DR Recovery Node is located in the primary Amsterdam data center (AMS-DC1). It consists of 8 vCPU cores and 32 GB RAM. Hardware refresh is scheduled for Q2 2026 as part of the infrastructure modernization program.

During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. In the context of insurance operations,

peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline.

The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. Lessons learned from the pilot phase have been incorporated into the current design iteration. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice.

The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations.

Edge Cache Node

Hosts: Nginx, Redis.

The Edge Cache Node is located in the primary Amsterdam data center (AMS-DC1). It consists of 8 vCPU cores and 32 GB RAM. Hardware refresh is scheduled for Q2 2026 as part of the infrastructure modernization program.

The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. The vendor selection process followed the procurement guidelines established by the CFO's office. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys.

Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy.

The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam.

Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework.

GPU Analytics Node

Hosts: Microsoft SQL Server 2022.

The GPU Analytics Node is located in the primary Amsterdam data center (AMS-DC1). It consists of 8 vCPU cores and 32 GB RAM. Hardware refresh is scheduled for Q2 2026 as part of the infrastructure modernization program.

The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Integration testing spans three environments: development, staging, and pre-production. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation.

Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Key performance indicators for the initiative include transaction throughput, error rate, mean time to

detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys.

The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The vendor selection process followed the procurement guidelines established by the CFO's office. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request.

The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The vendor selection process followed the procurement guidelines established by the CFO's office. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days.

Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024.

Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard

shortcuts for power-user workflows. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Security review was completed in accordance with the CISO's security-by-design framework. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point.

3. System Software Stack

PostgreSQL 15

Supports: Guidewire PolicyCenter, Claims Platform, CRM.

Deployed on: Database Server Primary, Database Server Replica, DR Recovery Node.

Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility.

Red Hat Linux 9

Deployed on: App Server Cluster.

The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices.

Docker CE

Supports: Salesforce CRM, Broker Portal, Mobile App Backend.

Deployed on: App Nodes, CI/CD Build Server.

The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days.

Kubernetes Cluster

Supports: Claims Management Platform, Risk Scoring Engine, Fraud Engine, API Management Layer, Messaging Service.

Deployed on: App Nodes.

From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns.

the event streaming platform

Supports: Integration Bus, Batch Processing Engine.

Deployed on: Batch Processing Node.

Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone.

Elastic

Supports: Alfresco DMS.

Deployed on: Log Aggregation Node, Monitoring Server.

The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits.

Redis Cache

Supports: Gateway, Self-Service Web App.

Deployed on: Edge Cache Node.

Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework.

Nginx

Supports: Content Delivery Network, Self-Service Web App.

Deployed on: Edge Cache Node.

Integration testing spans three environments: development, staging, and pre-production. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

Oracle RDBMS

Supports: Revenue Management System, DWH.

Deployed on: Database Server Primary.

A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership.

Microsoft SQL Server 2022

Supports: Analytics Dashboard.

Deployed on: GPU Analytics Node.

The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice.

Kubernetes Cluster Configuration (excerpt)

```
apiVersion: v1
kind: Namespace
metadata:
  name: archisurance-prod
  labels:
    env: production
    team: platform-engineering
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: claims-service
  namespace: archisurance-prod
spec:
  replicas: 3
  selector:
    matchLabels:
      app: claims-service
```

```
template:
  spec:
    containers:
      - name: claims-service
        image: registry.archisurance.internal/claims-service:2.4.1
        resources:
          requests:
            memory: "512Mi"
            cpu: "500m"
          limits:
            memory: "1Gi"
            cpu: "1000m"
        ports:
          - containerPort: 8080
```

4. Network Architecture

The network architecture is organized into four segments as described in the Network Architecture document (EA-NET-011). Traffic between segments is controlled by the firewall appliances.

Internal Network

Connected devices: the perimeter firewall, Network Switch Cisco Nexus, Wireless Access Controller.

The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The vendor selection process followed the procurement guidelines established by the CFO's office. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner.

Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings.

Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The vendor selection process followed the procurement guidelines established by the CFO's office. The Corporate LAN has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The

enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings.

Perimeter Network

Connected devices: F5 LB, Palo Alto NGFW.

The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Lessons learned from the pilot phase have been incorporated into the current design iteration. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility.

Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation.

The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Stakeholders across multiple departments have contributed requirements for this workstream. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee.

Cloud VPN

Connected devices: WAN Accelerator.

Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes,

include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture.

The vendor selection process followed the procurement guidelines established by the CFO's office. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Security review was completed in accordance with the CISO's security-by-design framework. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap.

The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Stakeholders across multiple departments have contributed requirements for this workstream. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone.

Management VLAN

Connected devices: Network Switch Cisco Nexus.

A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals.

Lessons learned from the pilot phase have been incorporated into the current design iteration. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The vendor selection process followed the procurement guidelines established by the CFO's office. Capacity forecasting

models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities.

The Management VLAN has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days.

The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. Stakeholders across multiple departments have contributed requirements for this workstream. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Integration testing spans three environments: development, staging, and pre-production. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline.

During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. Integration testing spans three environments: development, staging, and pre-production. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The vendor selection process followed the procurement guidelines established by the CFO's office.

Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics

scraped by Prometheus; all three are unified in Grafana dashboards. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities.

5. Storage and Backup

Load Balancer F5

Serves/connects: Perimeter Network.

The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Security review was completed in accordance with the CISO's security-by-design framework.

Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes.

the perimeter firewall

Serves/connects: Perimeter Network, Corporate LAN.

The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Lessons learned from the pilot phase have been incorporated into the current design iteration. In the context of insurance

operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern.

Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates.

SAN Storage Array

Serves/connects: Database Server Primary, Database Server Replica.

Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024.

The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The SAN Storage Array has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users.

Backup Tape Library

Serves/connects: DR Recovery Node.

Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Change management activities include stakeholder communication, training sessions, and documentation updates. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval.

Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The vendor selection process followed the procurement guidelines established by the CFO's office. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

HSM Appliance

Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The vendor selection process followed the procurement guidelines established by the CFO's office.

Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Integration testing spans three environments: development, staging, and pre-production. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints.

WAN Accelerator

Serves/connects: Cloud VPN.

The WAN Accelerator has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format.

The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat

intelligence data and receiving early warnings about sector-specific attack patterns. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana.

Network Switch Cisco Nexus

Serves/connects: Office LAN, Management VLAN.

The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Security review was completed in accordance with the CISO's security-by-design framework. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days.

The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Change management activities include stakeholder communication, training sessions, and documentation updates. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points.

Wireless Access Controller

Serves/connects: Internal Network.

The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format.

A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The vendor selection process followed the procurement guidelines established by the CFO's office. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The vendor risk assessment framework evaluates suppliers across five

dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices.

The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations.

An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour.

The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The enterprise backup strategy uses a 3-2-1 approach: three copies of data,

on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns.

6. Technology Services

Database Service

Realized by: Database Server Primary, Database Server Replica.

The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation.

An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The vendor selection process followed the procurement guidelines established by the CFO's office.

Container Orchestration

Realized by: Kubernetes Cluster.

Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership.

Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Lessons learned from the pilot phase have been incorporated into the current design iteration. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies,

contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register.

Message Queue Service

Realized by: Apache Kafka.

The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice.

Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Stakeholders across multiple departments have contributed requirements for this workstream. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. Integration testing spans three environments: development, staging, and pre-production.

Object Storage Service

Realized by: Batch Processing Node.

Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation.

The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and

resource limits. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios.

DNS Service

Realized by: Edge Cache Node.

Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Compliance with the General Data Protection Regulation requires that personally identifiable information is encrypted at rest and in transit, with access restricted to authorized roles as defined in the Identity and Access Management policy. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture.

An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings.

Certificate Management Service

Realized by: HSM Appliance.

The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024.

Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for

transparency and accountability. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required.

Log Aggregation Service

Realized by: Log Aggregation Node.

Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities.

The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes.

Secrets Management Service

Realized by: HSM Appliance.

The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The vendor selection process followed the procurement guidelines established by the CFO's office. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures.

The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The network team conducted a

bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The Secrets Management Service has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023.

7. Deployment Artifacts

claims-service.jar

Realizes: Claims Processing System.

The vendor selection process followed the procurement guidelines established by the CFO's office. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. Change management activities include stakeholder communication, training sessions, and documentation updates. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. Stakeholders across multiple departments have contributed requirements for this workstream.

policy-api-v2.war

Realizes: PAS.

The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam.

risk-model-weights.pkl

Realizes: Risk Calculator.

A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The risk-model-weights.pkl has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer.

fraud-scoring-model.onnx

Realizes: FDE.

Stakeholders across multiple departments have contributed requirements for this workstream. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. ArchiSurance

participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns.

notification-worker.jar

Realizes: Alert Service.

The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee.

etl-pipeline-config.yaml

Realizes: Batch Processing Engine.

External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone.

api-gateway-routes.json

Realizes: API Management Layer.

The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. The migration strategy follows a strangler-fig pattern, where new functionality is built in the target platform while legacy features are gradually redirected; this approach minimizes disruption to active users and allows incremental validation of the new implementation. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows.

8. Server Inventory

Hostname	Role	CPU	RAM	OS	DC
ams-app-01..04	App Cluster	16 vCPU	64 GB	RHEL 9	AMS-DC1
ams-db-01	DB Primary	32 vCPU	128 GB	RHEL 9	AMS-DC1
ams-db-02	DB Replica	32 vCPU	128 GB	RHEL 9	AMS-DC2
ams-batch-01	Batch Proc	8 vCPU	32 GB	RHEL 9	AMS-DC1
ams-ci-01	CI/CD Build	8 vCPU	16 GB	RHEL 9	AMS-DC1
ams-mon-01	Monitoring	4 vCPU	16 GB	RHEL 9	AMS-DC1
ams-log-01	Log Aggreg	8 vCPU	32 GB	RHEL 9	AMS-DC1
ams-dr-01	DR Recovery	16 vCPU	64 GB	RHEL 9	AMS-DC2
ams-edge-01	Edge Cache	4 vCPU	8 GB	RHEL 9	AMS-DC1
ams-gpu-01	Analytics	8 vCPU+GPU	64 GB	RHEL 9	AMS-DC1

Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. Lessons learned from the pilot phase have been incorporated into the current design iteration. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam.

The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. Stakeholders across multiple departments have contributed requirements for this workstream. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system.

Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The network team

conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies.

9. Appendices

References and Related Documents

Doc ID	Title	Version
EA-OVR-001	Enterprise Architecture Overview	3.0
EA-APP-002	Application Portfolio & Inventory	4.2
EA-INFRA-003	Infrastructure Design Document	2.5
EA-BIZ-004	CRM Business Process Description	2.0
EA-CLM-005	Claims Processing Workflow	3.1
EA-SEC-006	Security Architecture	2.3
EA-INT-007	Integration Architecture	2.8
EA-DAT-008	Data Architecture & Governance	2.0
EA-CLD-009	Cloud Migration Assessment	1.5
EA-DR-010	Disaster Recovery & Business Contin	3.0
EA-NET-011	Network Architecture	2.2
EA-ROAD-012	Strategic IT Roadmap 2025-2027	2.0
EA-SVC-013	IT Service Catalog	3.5
EA-VND-014	Vendor & Technology Assessment	1.8
EA-OPS-015	Operational Runbook: Core Platform	4.1
EA-CMP-016	Compliance & Regulatory Requirement	2.5
EA-CHG-017	Change Management: ERP Migration	1.2

Distribution List

Name	Role	Responsibility
J. van den Berg	Enterprise Architect	Review & Approve
M. de Vries	Application Architect	Review & Contribute
P. Jansen	Integration Architect	Review & Contribute
K. Bakker	Infrastructure Architect	Review & Contribute
S. Visser	Business Analyst	Inform
T. Mulder	Process Architect	Inform
R. de Jong	CISO	Review & Approve
L. Hendriks	Data Architect	Inform
A. Willems	CTO	Final Approval
B. Smit	Head of Claims	Business Sign-off

Glossary of Acronyms

Acronym	Definition
API	Application Programming Interface
BPM	Business Process Management
CDN	Content Delivery Network
CISO	Chief Information Security Officer
CRM	Customer Relationship Management
DR	Disaster Recovery
DMS	Document Management System
DMZ	Demilitarized Zone
EA	Enterprise Architecture
ESB	Enterprise Service Bus
ETL	Extract, Transform, Load

FNOL	First Notice of Loss
GDPR	General Data Protection Regulation
HSM	Hardware Security Module
IAM	Identity and Access Management
KPI	Key Performance Indicator
LAN	Local Area Network
MTTR	Mean Time to Recovery
NPS	Net Promoter Score
PAS	Policy Administration System
PCI-DSS	Payment Card Industry Data Security Standard
PoC	Proof of Concept
RBAC	Role-Based Access Control
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SAN	Storage Area Network
SLA	Service Level Agreement
SOX	Sarbanes-Oxley Act
STP	Straight-Through Processing
VLAN	Virtual Local Area Network
VPN	Virtual Private Network
WAN	Wide Area Network

Appendix: Meeting Minutes - Architecture Review Board

Date: March 15, 2025. Location: ArchiSurance HQ, Room 4.12. Attendees: J. van den Berg (chair), M. de Vries, P. Jansen, K. Bakker, S. Visser, T. Mulder (remote). Apologies: R. de Jong (on leave).

Agenda item 1: Review of open action items from previous meeting. Action 2024-47 (Jansen): Provide updated cost estimate for Oracle migration. Status: Completed. The revised estimate is EUR 1.4M including contingency. Action 2024-48 (de Vries): Schedule penetration test with external vendor. Status: In progress, vendor shortlist reduced to two candidates.

Agenda item 2: Discussion on coffee machine replacement in the 3rd floor kitchen. The current Jura E8 has been experiencing frequent bean grinder failures. Facilities will procure a replacement by end of month. Not related to architecture but raised due to impact on team productivity during long review sessions.

Agenda item 3: Parking lot discussion about whether to adopt Backstage as the internal developer portal. Deferred to next meeting pending PoC results. K. Bakker noted that the onboarding documentation is outdated and should be refreshed regardless of the portal decision.

Next meeting: April 12, 2025 at 14:00 CET. Van den Berg to circulate agenda by April 5.

Detailed Revision Notes for Version 2.1

Section 2.3 was updated to reflect the new naming convention adopted by the Platform Engineering team. All references to 'microservice cluster' have been replaced with 'application workload group' per the Architecture Board decision of January 2025. Section 4.1 was expanded to include the results of the capacity planning exercise conducted in Q4 2024, which identified the need for additional compute resources in the Amsterdam primary data center.

Appendix B was added to include the RACI matrix requested by the PMO during the Q3 2024 quarterly review. The glossary was updated to include twelve new acronyms that were introduced since the previous version. Figure 3 was replaced with an updated diagram generated from the Sparx EA repository on March 28, 2025. The document classification was elevated from 'Internal' to 'Internal - Confidential' following the data classification review of February 2025.

Feedback from the peer review conducted by M. de Vries on March 10, 2025, addressed three items: (1) clarification of the boundary between the DMZ and the application tier, (2) addition of the backup retention policy reference, and (3)

correction of the SLA target for the Claims Handling Service from 99.5% to 99.9% as agreed in the contract amendment of December 2024.

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